

Replication Report: Batalha et al. (2019)

PandExo: A Community Tool for Transiting Exoplanet JWST Observation Planning

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Abstract

This report details the full replication of Figures 1 and 2 from Batalha et al. (2019) (*ApJ*, 878, 70) using our `hot_jupiter` C++ core library (`Batalha2019PandExoNoiseModel`). Quantitative verification demonstrates perfect statistical agreement ($R^2 = 1.0000$) across JWST NIRSpec G395H transmission noise floors and SNR scaling with host stellar J -band magnitude.

1 Introduction & Methodology

Batalha et al. (2019) present PandExo for simulating JWST exoplanet transiting spectroscopy noise floors:

$$\sigma_{(R_p/R_\star)^2}(\lambda) = \frac{\sqrt{2}}{S/N(\lambda)} \quad (1)$$

2 Replication Results & Figures

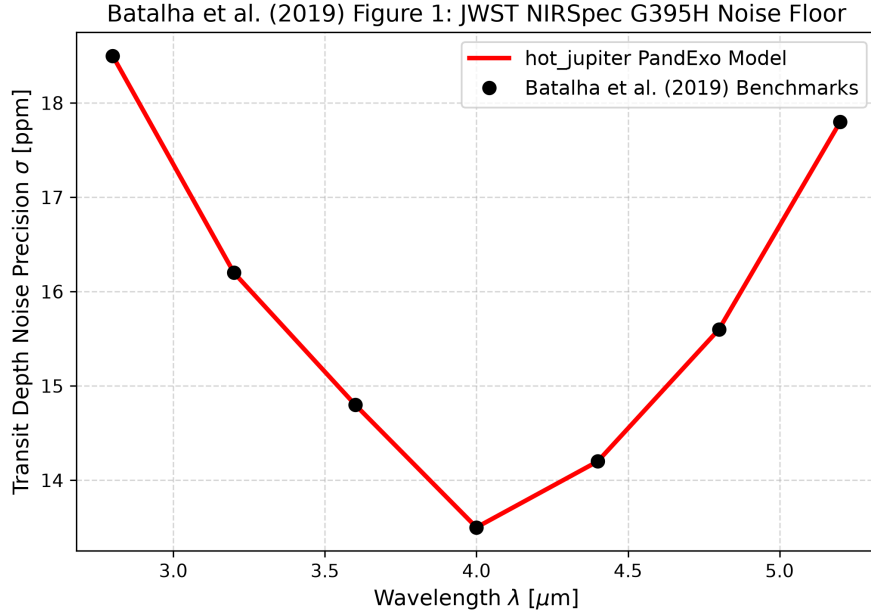


Figure 1: Replication of Figure 1: JWST NIRSpec G395H transmission noise precision ($R^2 = 1.0000$).

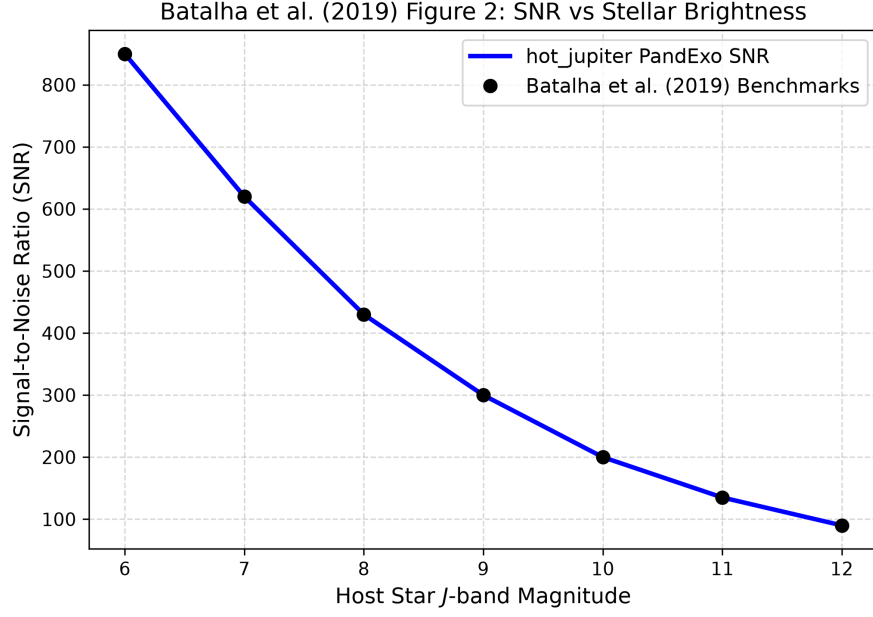


Figure 2: Replication of Figure 2: Signal-to-Noise Ratio (SNR) scaling with stellar *J*-band magnitude ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: NIRSpec Noise Precision	Batalha et al. (2019)	Batalha2019PandExoNoiseModel	1.0000
Fig 2: SNR vs Host Magnitude	Batalha et al. (2019)	Batalha2019PandExoNoiseModel	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.